HARD

GEOMETRY AND TRIGONOMETRY

Lines, angles, and triangles

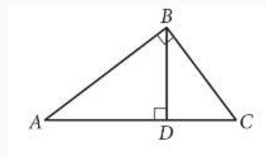
02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

In the figure above, $BD = 6$ and $AD = 8$. What is the length of $\overline{DC}$?

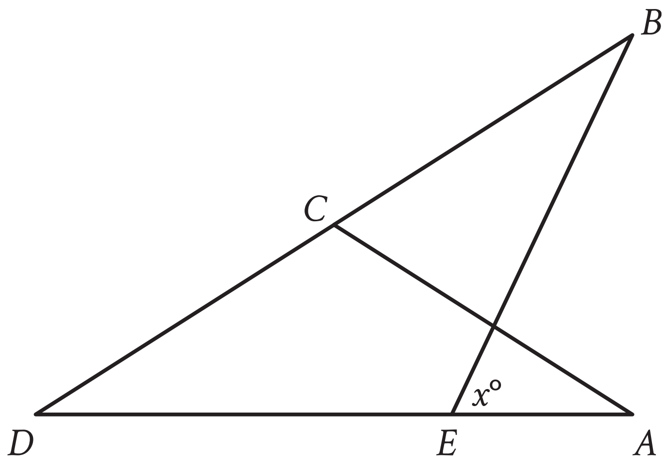
STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Triangle upper A upper C upper D partially overlaps triangle upper E upper B upper D.
- Point upper C is on line segment upper B upper D.
- Point upper E is on line segment upper A upper D.
- The measure of angle upper A upper E upper B is x°.
- A note indicates the figure is not drawn to scale.

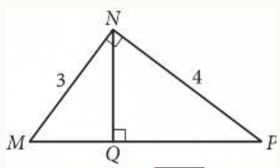
In the figure, $AC = CD$. The measure of angle EBC is $45°$, and the measure of angle ACD is $104°$. What is the value of x?

STUDENT-PRODUCED RESPONSE

6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 GEOMETRY AND TRIGONOMETRY



In the figure above, what is the length of $\overline{NQ}$?

A. 2.2

B. 2.3

C. 2.4

D. 2.5

Q4 GEOMETRY AND TRIGONOMETRY

In triangles ABC and DEF, angles B and E each have measure $27°$ and angles C and F each have measure $41°$. Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF?

A. The measure of angle A

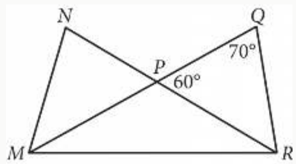
B. The length of side AB

C. The lengths of sides BC and EF

D. No additional information is necessary.

Q5

GRID-IN GEOMETRY AND TRIGONOMETRY



In the figure above, $\overline{MQ}$ and $\overline{NR}$ intersect at point P, $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

GEOMETRY AND TRIGONOMETRY

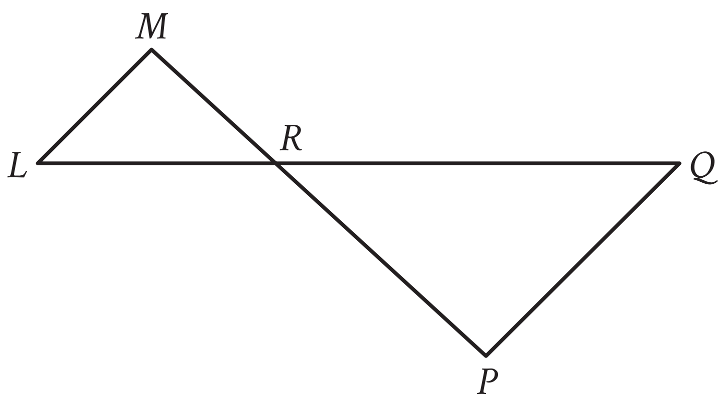
In triangle ABC, the measure of angle B is 90° and BD is an altitude of the triangle. The length of AB is 15 and the length of AC is 23 greater than the length of AB. What is the value of $\frac{BC}{BD}$?

A. $\frac{15}{38}$

B. $\frac{15}{23}$

C. $\frac{23}{15}$

D. $\frac{38}{15}$



Note: Figure not drawn to scale.

- Triangle upper L upper M upper R has a common vertex with triangle upper P upper Q upper R.
- The common vertex is upper R.
- A note indicates the figure is not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R, and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$, $\overline{LR}$, and $\overline{RP}$ are 6, 7, and 11, respectively. What is the length of $\overline{LQ}$?

A. $\frac{119}{11}$

B. $\frac{77}{6}$

C. $\frac{113}{6}$

D. $\frac{119}{6}$

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY

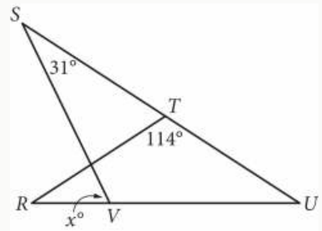
In right triangle ABC, angle C is the right angle and $BC = 162$. Point D on side AB is connected by a line segment with point E on side AC such that line segment DE is parallel to side BC and $CE = 2AE$. What is the length of line segment DE?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GEOMETRY AND TRIGONOMETRY



In the figure above, $RT = TU$. What is the value of x ?

A. 72

B. 66

C. 64

D. 58

Q10

GEOMETRY AND TRIGONOMETRY

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $9x - 140°$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

A. $-18x + 280°$

B. $-18x + 640°$

C. $18x - 280°$

D. $180°$